

Plausible, Scrutinized, Dismantled: A Rigorous Negative Result for Physiology-Constrained Cross-Hospital Transfer

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Abstract

Physiology-constrained pretraining does not improve cross-hospital sepsis prediction. A well-motivated hypothesis – that penalizing violations of known physiological relationships during masked-value pretraining of a clinical time-series transformer would yield representations that transfer better across hospitals than an unconstrained baseline – initially appeared confirmed: the constrained model beat empirical risk minimization (ERM) at every held-out site. Systematic re-evaluation traced the entire apparent gain to five identifiable evaluation confounds. Correcting all five reverses the result: the constrained model falls below ERM at every site, a plain gradient-boosted-tree baseline on aggregate features matches or beats both neural methods, and invariant risk minimization with 30 real hospital environments also fails to improve on ERM, ruling out an insufficient-environment explanation. Run-to-run seed variance of 4 to 7 AUROC points exceeds most of the originally reported effect sizes. The negative result also holds under a second, independently designed test: a label-free model-selection criterion built on the same physiological signal fails to reliably outperform a simple unsupervised baseline. We report each confound mechanistically and in generalizable terms, so a reader building a different domain-knowledge-constrained clinical method can recognize the same traps before publishing them.

Keywords: physiology-constrained learning, negative results, cross-hospital generalization, evaluation confounds, clinical machine learning, reproducibility

Data and Code Availability All experiments use three publicly available, de-identified, credentialed-access ICU databases: the PhysioNet/Computing in Cardiology Challenge 2019

dataset (Reyna et al., 2020), MIMIC-IV (Johnson et al., 2023), and the eICU Collaborative Research Database (Pollard et al., 2018). Code is not being made publicly available at this time.

Institutional Review Board (IRB) This work is a secondary analysis of pre-existing, de-identified, credentialed-access datasets already approved for secondary research use under their respective data use agreements. No new human-subjects data was collected, and no additional IRB approval was required.

1. Introduction

Clinical machine learning models trained at one hospital routinely lose accuracy when deployed at another, and the consequences are not hypothetical. An external validation of a proprietary sepsis prediction model widely deployed across US hospitals found an AUROC of 0.63, a sensitivity of 33%, and a positive predictive value of 12% at a single academic medical center – well below the vendor’s reported performance – contributing to the model’s deactivation at that site (Wong et al., 2021).

This paper tests a specific, principled response to that problem: physiology-constrained learning (PCL), which augments masked-value pretraining of a clinical time-series transformer with differentiable penalties for violating known physiological relationships among simultaneously measured variables. The premise follows the Independent Causal Mechanisms principle (Schölkopf et al., 2021): relationships such as the mean arterial pressure identity or the Henderson-Hasselbalch equation hold at every hospital regardless of local measurement device, analyzer calibration, or ordering practice, so a representation constrained to respect them should depend less on site-specific artifacts than one that is not. This is not a strawman hypothesis; it is the re-

sponse a careful practitioner reaches for when asked to build a clinical model that generalizes across sites, and it is the hypothesis an earlier version of this investigation set out to confirm.

Our contribution is not the claim that this hypothesis failed; it is a complete, rigorous diagnostic accounting of why a principled hypothesis produced a false positive convincing enough to submit, general enough that a reader building a different domain-knowledge-constrained method can recognize the same traps in their own pipeline before they reach publication. An earlier version of this investigation was submitted elsewhere and withdrawn after our own follow-up investigation uncovered the confounds this paper documents; we report that investigation in full here.

2. Method

PCL pretrains a six-layer clinical time-series transformer ($d=256$, 8 heads, ~ 3.2 M parameters) on 48-hour, hourly-binned ICU windows using a masked-value prediction objective, following the general masked-pretraining paradigm (Vaswani et al., 2017; Devlin et al., 2019). The training objective is $L = L_{\text{mask}} + \lambda L_{\text{phys}}$, where L_{phys} averages differentiable residuals on three physiological relations – an L_1 penalty on the mean arterial pressure identity and squared-error penalties on the Henderson-Hasselbalch and Severinghaus equations – applied only at timesteps where all required variables were observed and at least one was masked, so the penalty must be satisfied by imputation from cross-variable structure rather than by copying the input. Models are trained on PhysioNet Site A and evaluated zero-shot on PhysioNet Site B, MIMIC-IV, and eICU-CRD on the task of sepsis onset prediction.

3. The Initial, Misleading Result

Table 1 reports the original, first-pass result: PCL appears to beat every baseline at every held-out site, with the margin growing with distributional distance – +6.4 percentage points at Site B, +10.1 at MIMIC-IV, +8.4 at eICU – and PCL is the best method in every one of three seeds at every site. The result also passed a constraint-randomization ablation (real physiological constraints outperformed shuffled, wrong-equation,

and noise-injected variants, with the two most-corrupted variants falling below unconstrained ERM) and a measurement-noise robustness check (PCL’s AUROC degraded less than ERM’s under injected sensor bias). Nothing about the result looked fragile from any angle tried at the time; that is precisely what makes the diagnosis in Section 4 worth reporting in full.

Table 1: The original, uncorrected result: zero-shot cross-hospital sepsis AUROC, mean over 3 seeds. PCL appears to beat every baseline at every held-out site. This result does not survive Section 4 and is reported here only to show how convincing it looked.

Method	Source	Site B	MIMIC-IV	eICU
ERM	0.823	0.697	0.582	0.541
DRO	0.778	0.683	0.612	0.604
IRM [†]	—	0.675	—	—
PCL	0.865	0.761	0.682	0.626

[[†]IRM: single seed, Site B only, $k=3$ care-unit environments. Margins as originally reported: +6.4pp Site B, +10.1pp MIMIC-IV, +8.4pp eICU.]

4. The Five Confounds

Each confound below is stated mechanistically and in terms general enough that a reader building a different domain-knowledge-constrained method can check their own pipeline against it; each ends with the generalized lesson.

4.1. Label-Definition Shift

Training labels followed the clinical Sepsis-3 consensus definition (Singer et al., 2016), while external evaluation on MIMIC-IV and eICU originally used administrative ICD-code-derived sepsis labels. Clinical and claims-based sepsis definitions are known to diverge substantially (Rhee et al., 2017), so cross-site AUROC differences conflated genuine distribution shift with a change in the definition of the outcome itself. We corrected this by re-deriving Sepsis-3 directly via SOFA scoring anchored to a suspected-infection

window, following the standard reference implementation (Seymour et al., 2016), yielding sepsis prevalence of 30.9% on MIMIC-IV and 8.8% on eICU. **Lesson:** any cross-site comparison must confirm the outcome definition, not only the input distribution, is held fixed across sites.

4.2. Pretraining Data Leakage Disguised as Zero-Shot Transfer

PCL’s masked-value pretraining used unlabeled data from all four sites, while ERM’s pretraining used source-site data only. Although pretraining is self-supervised and touches no labels, exposure to target-site inputs during pretraining still breaks the zero-shot premise the comparison depends on: PCL had, in effect, already seen the deployment distribution. **Lesson:** “no labels touched” is not the same guarantee as “no target data touched”; a zero-shot claim must audit every stage of training, not only the labeled fine-tuning stage.

4.3. OOD-Contaminated Hyperparameter Selection

The constraint weight λ , which strongly affects cross-hospital performance, was originally chosen using AUROC on a held-out target site rather than on source-domain validation data, so the reported zero-shot numbers were selected with knowledge of the test distribution they were meant to predict. **Lesson:** a hyperparameter is only as zero-shot as the data used to select it; auditing final-model performance is not sufficient if the selection procedure that produced that model touched target data.

4.4. Partial Circularity in the Severinghaus Constraint

PhysioNet does not record arterial oxygen partial pressure (PaO₂) directly; the released dataset reconstructs it from oxygen saturation by inverting the Severinghaus equation – the same relation PCL’s constraint penalizes the model for violating. The constraint’s training-time supervision was therefore partly self-referential: it could be satisfied, in part, by recovering a value the data-generating process had already computed from the constraint’s own equation. **Lesson:** before enforcing a domain-knowledge constraint, audit

whether any of its input variables were themselves derived by assuming that constraint; a pre-processing pipeline can silently manufacture the invariance a model is later credited with discovering.

4.5. Measurement-Composition Artifact in the Aggregate Violation Score

The aggregate physiological violation score averages whichever of the three constraints happen to be computable at a given timestep, and constraint availability is highly uneven and site-dependent – for instance, bicarbonate is charted in 8.1% of Site A stay-hours versus 0.16% at Site B, a fifty-fold difference driven by local lab-ordering practice, not physiology. Because the sparser constraints also occupy a different residual scale than the dense ones, a site’s aggregate mean partly tracks which labs are ordered there rather than any physiological difference – reproducible on raw data with no model involved. **Lesson:** an aggregate score over heterogeneously-available components is a composition statistic before it is anything else; decompose it before reading a cross-site difference in it as a physiological effect.

Formal automated detectors for these five failure modes are the subject of a separate, in-progress companion paper by the same author, not yet complete or peer-reviewed; we do not cite it as an established tool here.

5. The Corrected Result

With all five confounds addressed, PCL falls below ERM at every site (Table 2), alongside a group-distributionally-robust baseline (Sagawa et al., 2020). Two further findings support that this is a genuine negative result rather than an artifact of the correction itself. First, a gradient-boosted tree (Chen and Guestrin, 2016) trained on simple per-stay summary statistics matches or exceeds both neural methods at every external site (0.754 / 0.654 / 0.660 versus ERM’s 0.701 / 0.649 / 0.592 and PCL’s 0.655 / 0.622 / 0.572), which further questions whether the transformer architecture is earning its complexity here. Second, invariant risk minimization (Arjovsky et al., 2019) with 30 real hospital environments drawn from eICU, evaluated

in-distribution within eICU rather than zero-shot, also fails to beat its own ERM baseline (0.788 versus 0.792) – ruling out an insufficient-environment-count explanation for IRM’s known low- k failure regime (Rosenfeld et al., 2021), on the same corrected pipeline; this in-distribution comparison uses its own ERM baseline and is not directly comparable to this table’s zero-shot eICU row. Finally, run-to-run variability across seeds reached 4 to 7 AUROC points on this pipeline, exceeding most of the differences originally reported as method effects in Table 1: a single-seed comparison at this scale is not informative regardless of how large the claimed gain looks.

Table 2: Corrected cross-hospital sepsis AUROC, mean over 3 seeds, all five confounds addressed. PCL is now below ERM at every site.

Method	Source	Site B	MIMIC-IV	eICU
ERM	0.818	0.701	0.649	0.592
PCL	0.798	0.655	0.622	0.572
Group DRO	0.822	0.717	0.688	0.628
XGBoost	—	0.754	0.654	0.660

6. Beyond Prediction Accuracy

The same physiological signal was also tested as a label-free criterion for selecting the constraint weight without target labels – an independently designed test, since a selection criterion can fail even when the underlying representation does not, and vice versa. In prior work by the same author, reconstruction error on unlabeled target data reduced mean selection regret from 0.052 to 0.012 relative to source-domain validation, while the physiology-violation criterion evaluated alongside it did not consistently outperform this simple unsupervised baseline (Tamvada and Lin, 2026). For the full selection-criterion comparison and the validation-failure diagnosis motivating it, see that work; we do not reproduce its tables here.

A calibration or conformal-prediction-coverage test was considered but is not included here: no such artifact was located in this codebase at the

time of writing, and we chose not to report a number we could not verify.

7. Limitations

This is a case study in how one principled domain-knowledge constraint failed on one task and one architecture; it is not a universal claim that domain-knowledge constraints cannot improve cross-hospital transfer. All experiments use a single clinical task (sepsis onset prediction) and a single transformer-based architecture and pre-training objective; whether the same confounds recur for other tasks, other architectures, or other domain-knowledge constraints is untested. Three random seeds is a small sample for a variance estimate, though the seed-variance finding itself (Section 5) is the point of that measurement, not a caveat to it.

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